CS & IT ENGINEERING

Digital Logic

Sequential Circuit

Lecture No. [redacted] 6

By- Aditya sir

Recap of Previous Lecture

Topic

Master Slave FF

D FF

T FF

Design of FF

Topics to be Covered

Topic

Sequential Circuit

- Design
- Register

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

@ADITYASIR_PW

Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW available ⟶ desired

Designing of Flip-Flop:

Step 1: → Write the characteristic Table of desired FF. CD

Step 2: → Write the excitation table of available FF. EA

Step 3: → Write the logical expression.

Step 4: → Minimize the logical expression.

Step 5: → Hardware Implementation.

#Q. Design a w-FF by using SR FF ?

Step 1

Step 2

W	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	0	1
1	0	1	1	0
0	1	1	X	0

Step 3

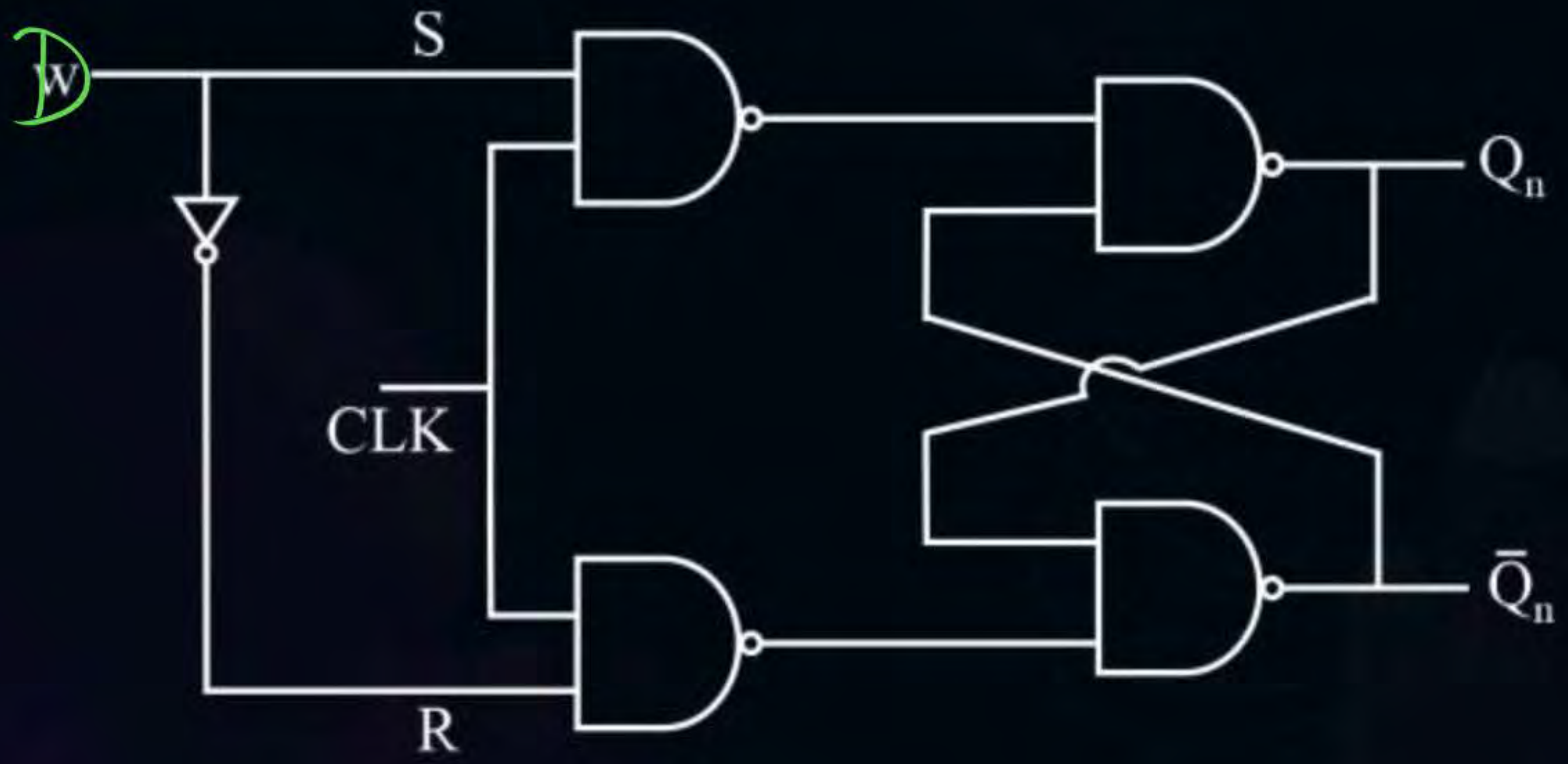
W \ Q_n	0	1
0		
1	1	x

Step 4

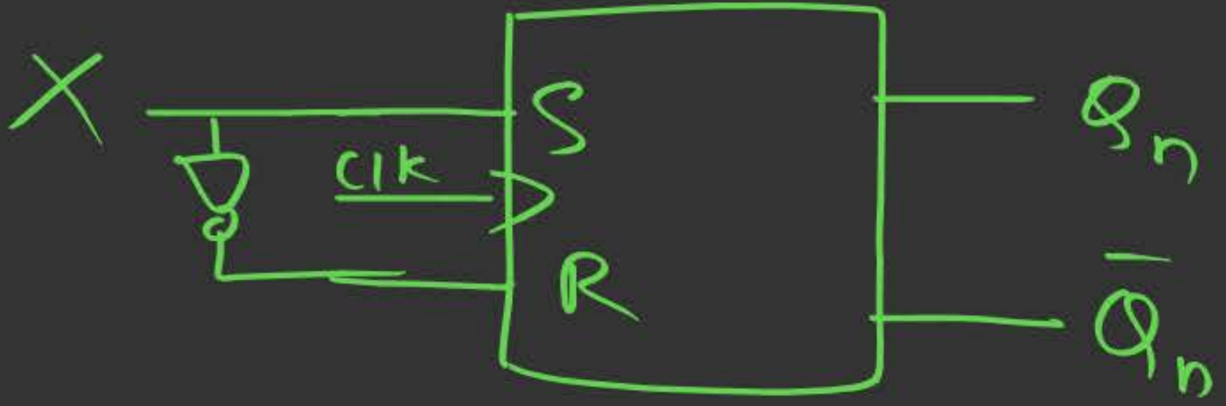
W \ Q_n	0	1
0	$\cancel{1}$	x
1		

Step 5

Topic : Sequential Circuit



Q.1 Imp :-



$S = X, \; R = \bar{X}$

$$Q_{n+1} = S + \bar{R} \cdot Q_n$$

$$= X + \bar{\bar{X}} \cdot Q_n$$

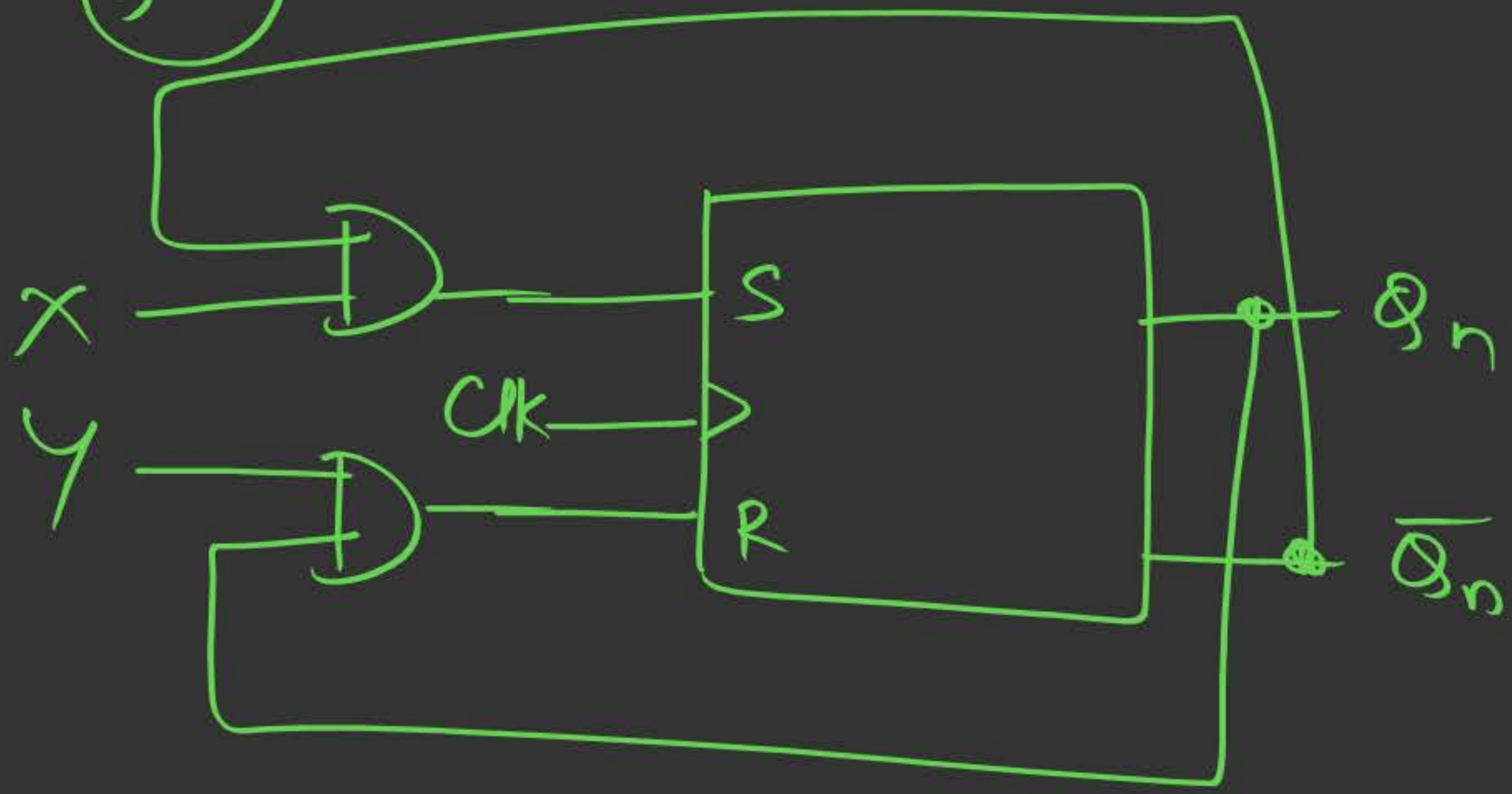
$$= X + X Q_n$$

$$= X(1 + Q_n)$$

$$= X$$

$$\boxed{Q_{n+1} = D} \Leftarrow \boxed{Q_{n+1} = X}$$

Q.2



$$S = X \cdot \overline{Q_n}, \quad R = Y \cdot Q_n$$

$$Q_{n+1} = S + \overline{R}\, Q_n$$

$$= X\overline{Q_n} + \overline{YQ_n} \cdot Q_n$$

$$= X\overline{Q_n} + (\overline{Y} + \overline{Q_n})\, Q_n$$

$$\boxed{Q_{n+1} = X\overline{Q_n} + \overline{Y}\, Q_n}$$

JK FF: $Q_{n+1} = J\overline{Q_n} + \overline{K}\, Q_n$

$$\boxed{X = J, \; Y = K}$$

this Sunday

$\boxed{12:30 - 3}$

✓

4:30 AM

12 AM

6 months

Q.) Design a T FF using SR FF

[NAT]

#Q. Design a T-FF by using SR FF ?

Step 1 →

Step 2

Desired → T FF

Avail → SR FF

T	Q_n	Q_{n+1}	S	R	
0	0	0	0	X	0
0	1	1	X	0	1
1	0	1	1	0	2
1	1	0	0	1	3

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Step 3

$S = \sum m(2) + \sum d(1)$

$R = \sum m(3) + d(0)$

$S \Rightarrow$

T \ Q_n	$\bar{Q}_n$ 0	Q_n 1
$\bar{T}$ 0	0	x 1
T 1	$\boxed{1}$ 2	3

$S = T\bar{Q}_n$

Step 4

$R \Rightarrow$

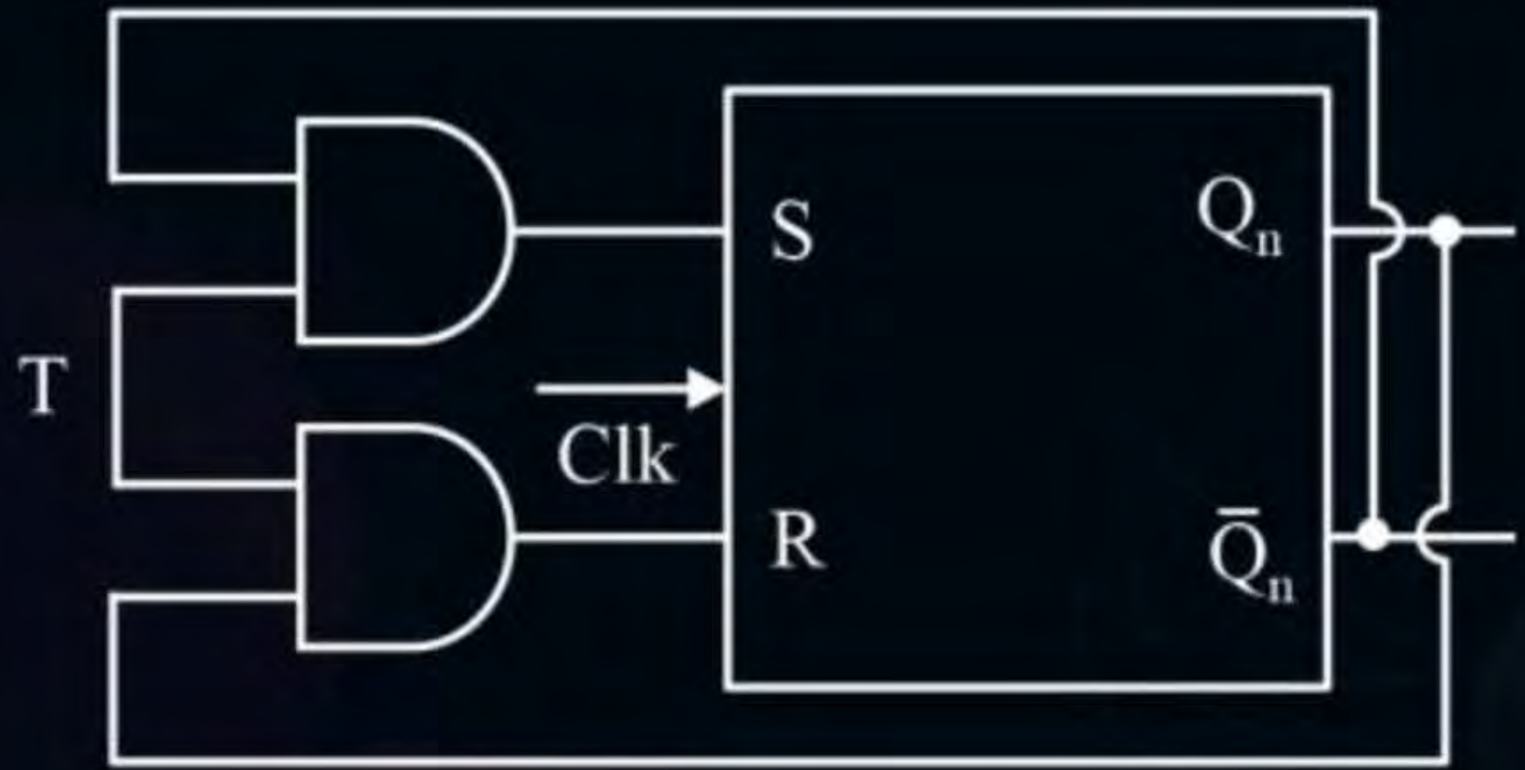
T \ Q_n	$\bar{Q}_n$ 0	Q_n 1
$\bar{T}$ 0	x 0	1
T 1	2	$\boxed{1}$ 3

$R = TQ_n$

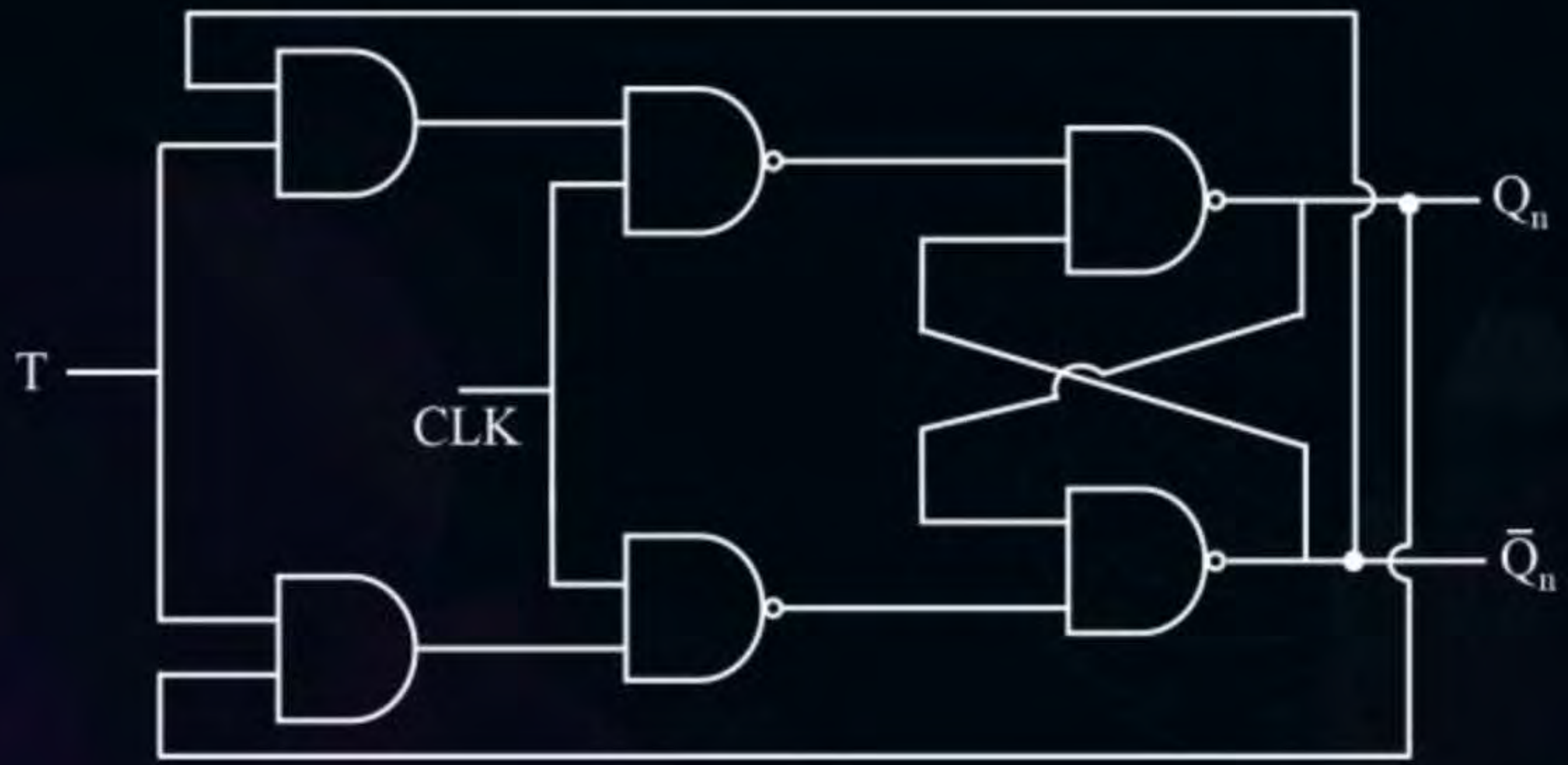
Step 5

$S = T\overline{Q_n}$

$R = TQ_n$



Topic : Sequential Circuit



Remember:-

$$\underbrace{SR \quad JK \quad D \quad T}_{12 \ \leftarrow\ 3\times 4}$$

SR

JK

D

T

(Q) Design T FF using D FF

[NAT]

#Q. Design a T-FF by using ~~W~~ D - FF ?

$D = Q_{n+1}$

Step 1

Step 2

$D(T, Q_n)$

T	Q_n	Q_{n+1}	~~W~~ D	
0	0	0	0	0
0	1	1	1	1
1	0	1	1	2
1	1	0	0	3

Step 3 & Step 4

~~W~~ D $= T \oplus Qn$

Step 5

$$D = T \oplus Q_n$$

H.W Challenge: Design a AJ FF using SR FF?

Truth Table

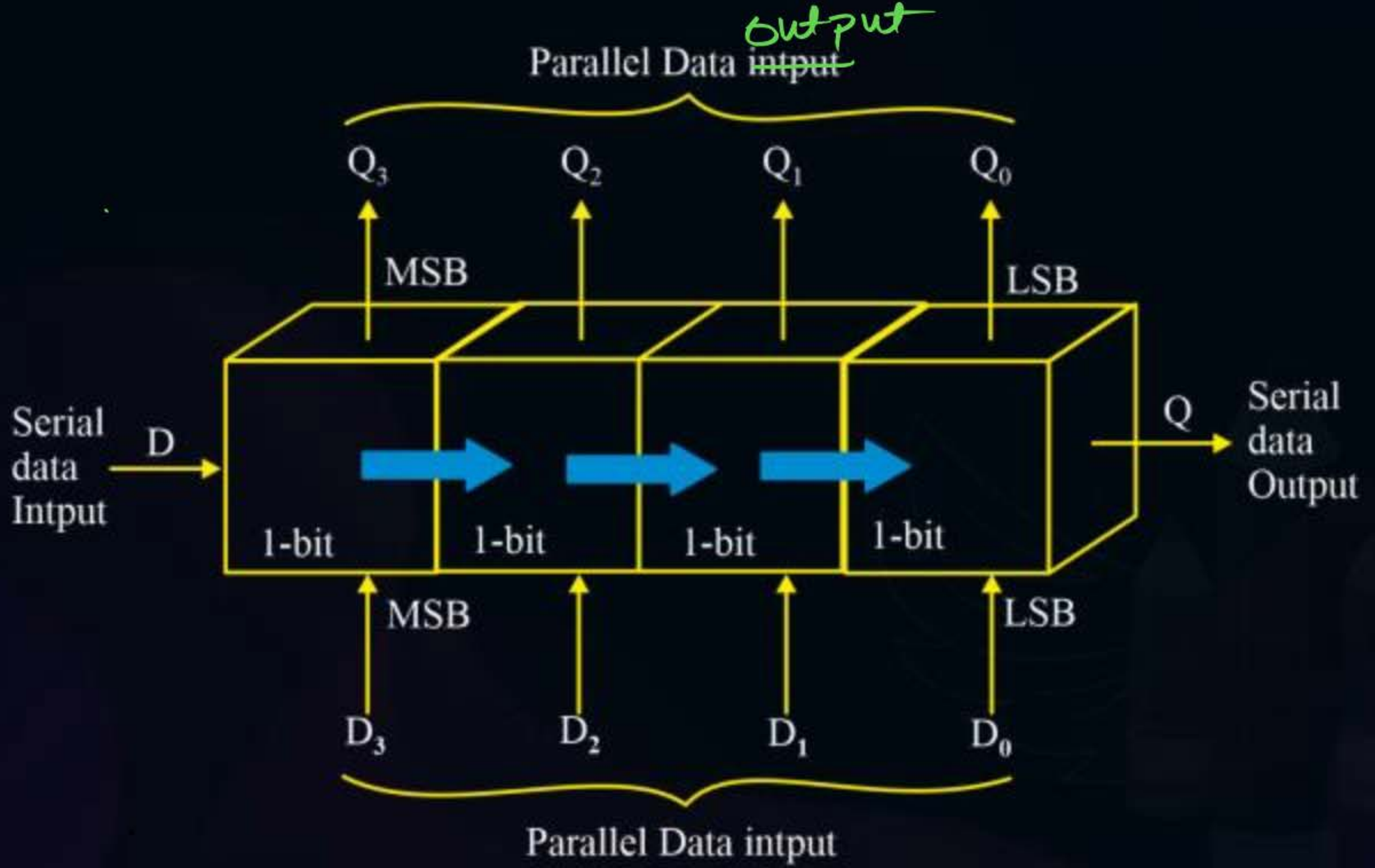
A	J	Q_{n+1}
0	0	$\overline{Q_n}$
0	1	Q_n
1	0	1
1	1	0

Register

Topic : Shift Register

1. Registers are used to store group of bits
2. To store "n" bits minimum "n" Flip Flips are required
3. Generally D Flip Flops are used to Design Register

Topic : Shift Register



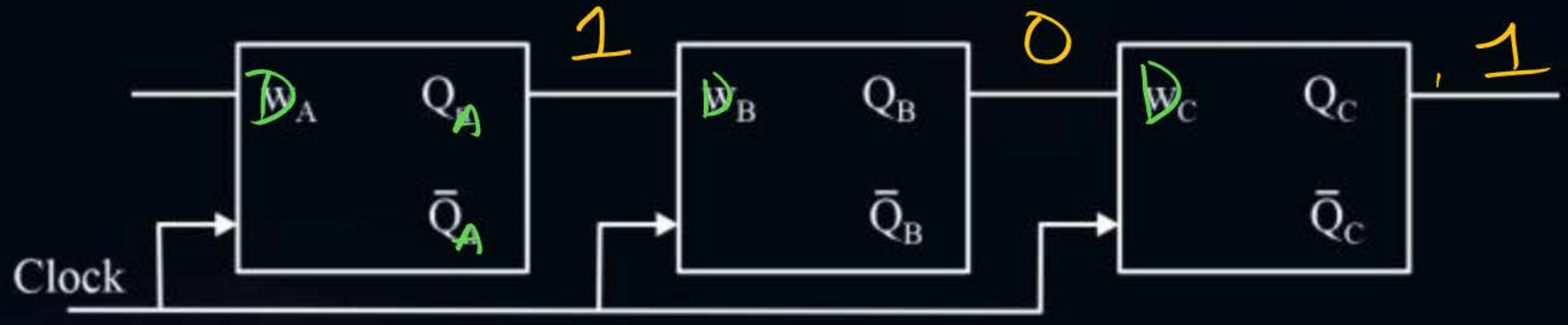
I/P → 2
O/P → 2

Topic : Shift Register

1. Serial input serial output shift register (SISO)
2. Serial input parallel output shift register (SIPO)
3. Parallel input serial output shift register (PISO)
4. Parallel input parallel output shift register (PIPO)

2 steps :- Store + Retrieve

Topic : SERIAL INPUT SERIAL OUTPUT (SISO) SHIFT REGISTER



initial

Clock	Input	Q_A	Q_B	Q_c
0	101	0	0	0
1		1	0	0
2		0	1	0
3		1	0	1

Topic : SERIAL INPUT SERIAL OUTPUT (SISO) SHIFT REGISTER

1. To store 'n' bits in 'n' bit SISO minimum 'n' clocks are required.
2. To retrieve 'n' bits from 'n' bit SISO minimum 'n - 1' clocks are required.
3. It is slowest shift register among all the shift registers.

Serial → Slow

Parallel → Fast

Total Clocks $= n + (n-1)$

$= 2n - 1$

Topic : SERIAL INPUT Parallel OUTPUT (SIPO) SHIFT REGISTER

SIPO

$\boxed{1|0|1}$

Clock	Input	Q_A	Q_B	Q_c
0	101	0	0	0
1		1	0	0
2		0	1	0
3		1	0	1

Topic : Sequential Circuit

- To store 'n' bit in 'n' bit SIPO minimum 'n' clocks are required.
- To retrieve 'n' bits from 'n' bit SIPO no clock required.

Total = n + 0
= n

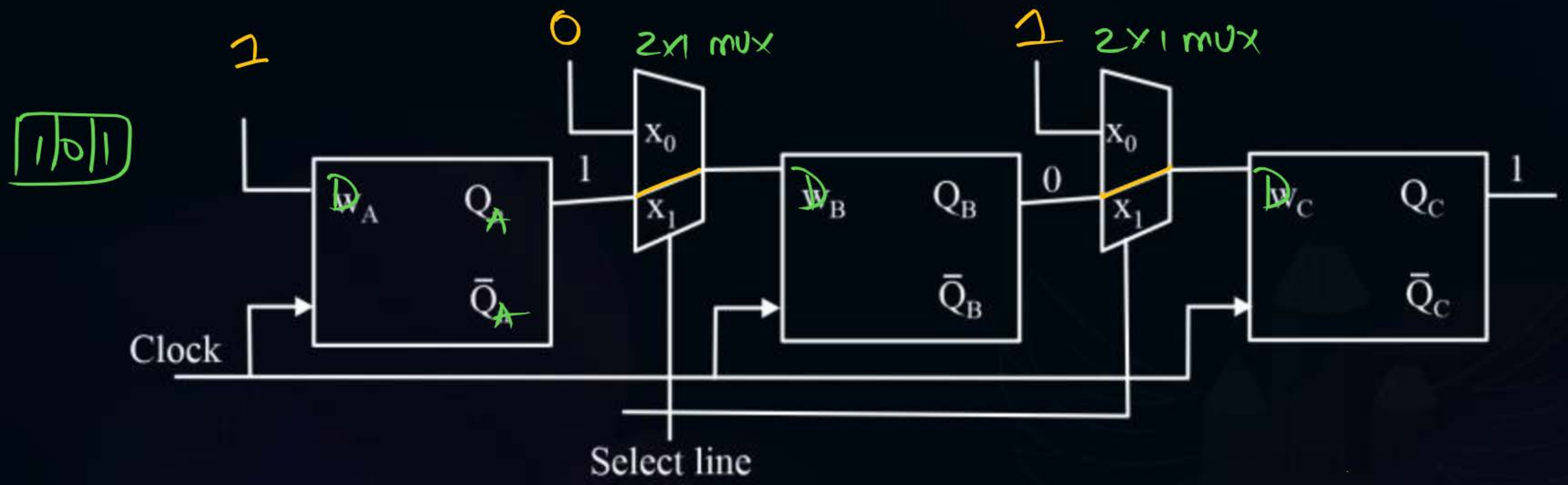
SIPO
(Temporal → Spatial)

Serial
Slow
(Temporal code)

Parallel
Fast
(Spatial code)

Topic : PISO SHIFT REGISTER

Parallel i/p , Serial o/p



Topic : PISO SHIFT REGISTER

1. To store 'n' bits in 'n' PISO only one clock is required.
2. To retrieve 'n' bits from 'n' bit PISO minimum (n - 1) clocks are required.

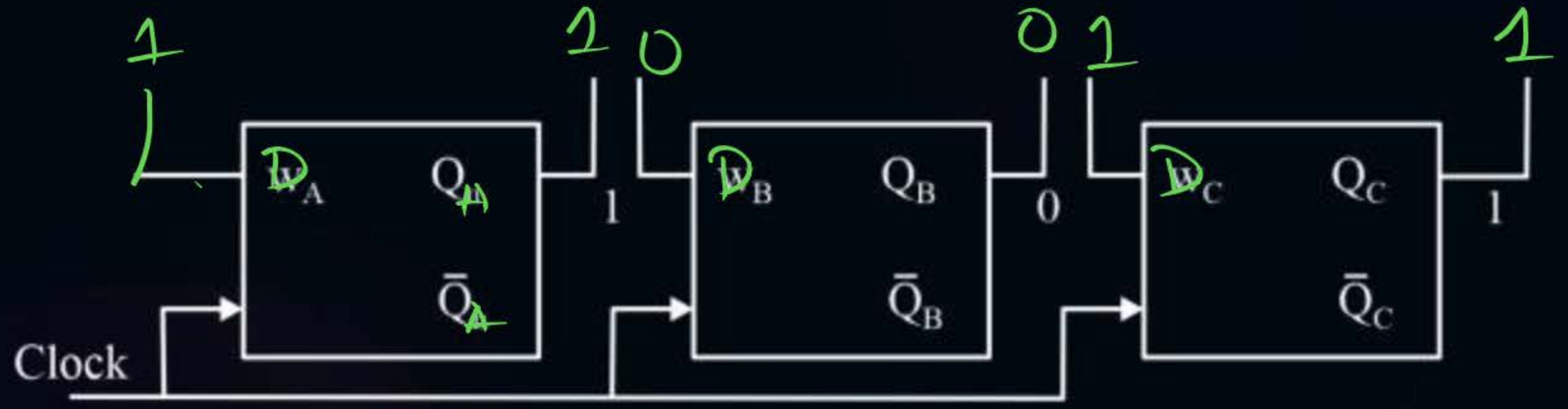
Total clocks = 1 + (n-1)

= n

PISO (spatial → Temporal)

Topic : PARALLEL INPUT PARALLEL OUTPUT (PIPO) SHIFT REGISTER

$\boxed{1|0|1}$



clock → 1

Store → 1 clk

Retrieval → 0 clk

Topic : Sequential Circuit

- To store 'n' bit in 'n' bit PIPO only one clock is required.
- There is no clock requirement in PIPO to retrieve 'n' bits.
- $\boxed{\text{Fastest}}$ shift register among all the shift registers.

PIPO → Fastest

SISO → slowest

Total clocks

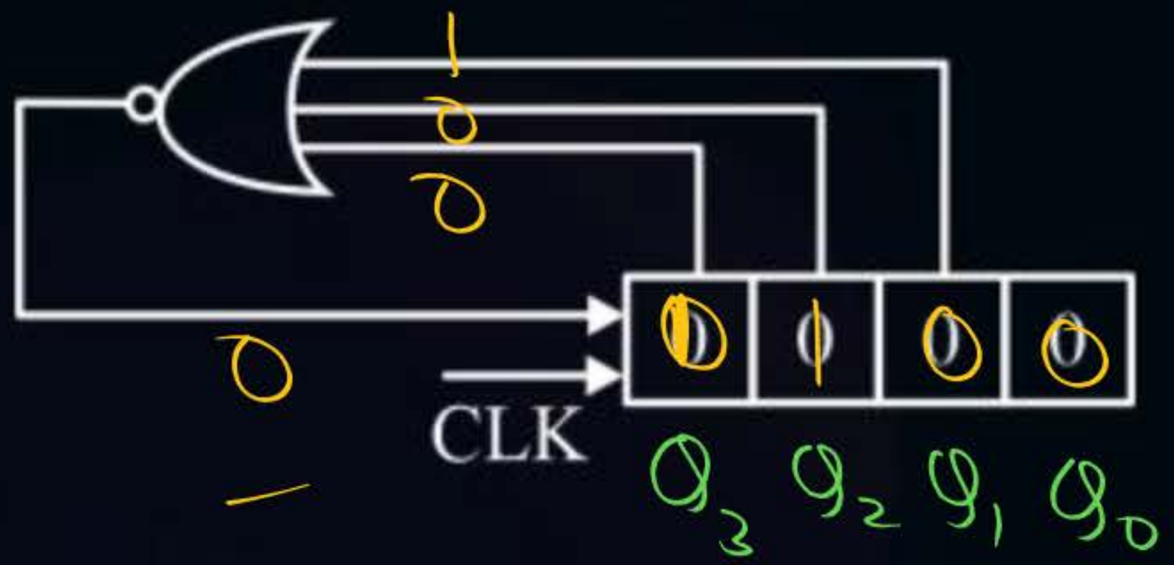
= 1+0

= 1

[MCQ]

#Q. Write the state of the register given below?

(Assume initially all flips flops are reset)



Clock	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1
5	1	0	0	0
6	0	1	0	0
7	0	0	1	0
8	0	0	0	1

H.W

1) what is the o/p after 50^{th} clock?

2) " " " " 105^{th} clock?

" " " " 1023^{th} clock)

Topic : Sequential Circuit

	Store	Relative	Total	
SISO	n	n-1	2n-1	→ Slowest
SIPO	n	0	n	
PISO	1	n-1	n	
PIPO	1	0	1	→ Fastest

2 mins Summary

→ Flip Flops

→ Registers

THANK - YOU